

"Falcon 2.0T: Camshaft over-advanced from sticking variable valve timing solenoid"

Document: tsb-23-224

Area: bulletins

1 Condition

Affected Falcon 2.0T vehicles may set DTC P0011 (Intake Camshaft Position Timing Over-Advanced or System Performance, Bank 1) accompanied by a rough idle, hesitation under light load, and an illuminated MIL. The condition typically appears after a cold start and may clear intermittently as oil temperature rises. Drivers may also report a momentary lack of power during the first acceleration event. Confirm the freeze-frame data and verify P0011 is current before proceeding, using the diagnostics described in the Falcon Engine Service Manual :: Variable Valve Timing System section.

2 Cause

Internal varnish accumulation can cause the VVT-3360 intake camshaft control solenoid spool to stick in a partially open position, allowing excess oil pressure to advance the camshaft beyond the commanded angle. The over-advanced condition is detected by the camshaft position correlation logic, which sets P0011. The VVT-3360 solenoid is the root-cause component and should not be cleaned and reused once sticking is confirmed. The complete operating theory is covered in the Falcon Engine Service Manual .

3 Repair Procedure

Replace the VVT-3360 intake camshaft control solenoid following PROC-VVT-SOL, including the O-ring renewal and torque-to-spec of the retaining bolt. Clear DTC P0011, perform the camshaft relearn, and road test through at least two cold-to-warm cycles to confirm the code does not return. Inspect the oil and filter condition during the repair, as degraded oil contributes to solenoid sticking. The detailed removal and installation sequence is given in the Falcon Engine Service Manual :: VVT Solenoid Replacement section.

4 Parts Information

Order one VVT-3360 intake camshaft control solenoid per affected bank; the kit ships with a new spool-bore O-ring that must be installed dry. Do not substitute the exhaust-side solenoid, which uses a different flow rate and connector keying. Verify the current supersession for VVT-3360 before ordering, as later production carries a revised filter-screen design. Torque values and fastener part numbers are listed in the Falcon Engine Service Manual :: VVT Solenoid Replacement reference.